

Recall from last class:

- **Theorem 10:** Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then there exists a unique matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$; namely $A = [T(\vec{e}_1) \ \dots \ T(\vec{e}_n)]$.
- **Theorem 11:** Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then T is injective if and only if $T(\vec{x}) = \vec{0}$ has only the trivial solution.
- **Theorem 12:** Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear, and let A be the associated matrix. Then
 1. T is surjective if and only if the columns of A span $\mathbb{R}^m$.
 2. T is injective if and only if the columns of A are linearly independent.

Let $A = [\vec{a}_1 \ \dots \ \vec{a}_n]$ be an m by n matrix. We express A by

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

where a_{ij} is the entry in the i -th row and j -th column.

Theorem: Let A , B , and C be matrices of the same size, and let r and s be real numbers. Then the following properties hold:

1. $A + B = B + A$.
2. $(A + B) + C = A + (B + C)$.
3. $A + 0 = A$.
4. $r(A + B) = rA + rB$.
5. $(r + s)A = rA + sA$.
6. $r(sA) = rsA$.

Definition 1. If A is a $m \times n$ matrix and B is an $n \times p$ matrix with columns $\vec{b}_1, \dots, \vec{b}_p$, then AB is the $m \times p$ matrix given by

$$AB = A[\vec{b}_1 \ \dots \ \vec{b}_p] = [A\vec{b}_1 \ \dots \ A\vec{b}_p].$$

1. Compute AB when

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}.$$

Can you also compute BA ?

Remark:

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Properties of Matrix Multiplication.

Theorem:

- (a) $A(BC) = (AB)C$.
- (b) $A(B + C) = AB + AC$.
- (c) $(B + C)A = BA + CA$.
- (d) $r(AB) = (rA)B = A(rB)$.
- (e) $I_m A = A = A I_n$.

2. True or False Questions. Prove or give a counterexample as appropriate. Suppose A and B are n by n matrices.

(a) $AB = BA$.

(b) If $AB = 0$, then $A = 0$ or $B = 0$.

(c) If $A^2 = 0$, then $A = 0$.

(d) If $AB = AC$, then $B = C$.

Definition 2. Given an $m \times n$ matrix A , then the **transpose of A** , denoted A^T is the $n \times m$ matrix given by taking the columns of A as rows.

Theorem: Let A be an m by n matrix, B be an n by p matrix, and $r \in \mathbb{R}$.

- (a) $(A^T)^T = A$.
- (b) $(A + B)^T = A^T + B^T$.
- (c) $(rA)^T = rA^T$.
- (d) $(AB)^T = B^T A^T$.

Definition 3. An n by n matrix A is **invertible** if there is an n by n matrix C so that $AC = I$ and $CA = I$. We denote the inverse of A by A^{-1} .

Natural Questions:

- (a) Do inverses always exist?
 - (b) If not, when do they exist?
 - (c) If they exist, are they unique?
 - (d) How can we find them?
3. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Determine when A is invertible and find A^{-1} .

Theorem: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Definition 4. We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by

$$\det(A) = ad - bc.$$

Theorem: If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.

4. Prove the Theorem.

Theorem: Suppose A and B are n by n matrices, then the following hold.

- (a) If A is invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- (b) $(AB)^{-1} = B^{-1}A^{-1}$.
- (c) $(A^T)^{-1} = (A^{-1})^T$.

5. Prove the Theorem.

Matrix Operations – Answers / Solutions

1. Let

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix},$$

then using the definition of matrix multiplication we get the following

$$A\vec{b}_1 = \begin{bmatrix} 2(4) + 3(1) \\ 1(4) - 5(1) \end{bmatrix} = \begin{bmatrix} 11 \\ -1 \end{bmatrix}, A\vec{b}_2 = \begin{bmatrix} 2(3) + 3(-2) \\ 1(3) - 5(-2) \end{bmatrix} = \begin{bmatrix} 0 \\ 13 \end{bmatrix}, \text{ and } A\vec{b}_3 = \begin{bmatrix} 2(6) + 3(3) \\ 1(6) - 5(3) \end{bmatrix} = \begin{bmatrix} 21 \\ -9 \end{bmatrix}.$$

Hence

$$AB = [A\vec{b}_1 \ A\vec{b}_2 \ A\vec{b}_3] = \begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{bmatrix}.$$

We cannot multiply the matrices in the other order as the number of rows of A does not equal the number of columns of B , and hence BA . In terms of the corresponding linear transformations, this says the number of outputs of A does not match the number of inputs of B , and hence we cannot compose the functions in that order.

2. (a) **False.** There are many counterexamples, but take for instance $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, and $B =$

$$\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}, \text{ then } AB = \begin{bmatrix} -2 & 3 \\ -4 & 7 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 3 & 4 \\ 2 & 2 \end{bmatrix}. \text{ Thus } AB \neq BA.$$

(b) **False.** Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, and $B = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$, then $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

(c) **False.** Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

(d) **False.** Let

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \text{ and } C = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix},$$

then $AB = AC$ but $B \neq C$.

3. We want to find e, f, g and h such that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus we get the following system of equations:

$$ae + gb = 1$$

$$af + bh = 0$$

$$ce + dg = 0$$

$$cf + dh = 1$$

Multiplying the first equation by c and the third equation by $-a$ and adding them together yields: $bcg - adg = c$, and assuming $ad - bc \neq 0$, we get $g = \frac{-c}{ad-bc}$.

Continuing in a similar manner we get $e = \frac{d}{ad-bc}$, $h = \frac{a}{ad-bc}$, and $f = \frac{-b}{ad-bc}$.

Thus the matrix is invertible if and only if $ad - bc \neq 0$, and we have an explicit formula for the inverse matrix.

4. Since A is invertible, we know A^{-1} exists. Let $\vec{x} = A^{-1}\vec{b}$, then we check $\vec{x}$ is a solution. Here $A\vec{x} = AA^{-1}\vec{b} = \vec{b}$, and hence $\vec{x}$ is a solution.

Now we check it is the unique solution. Suppose $\vec{u}$ is another solution, then $A\vec{u} = \vec{b}$. Since A is invertible, we can multiply both sides by A^{-1} to get $A^{-1}A\vec{u} = A^{-1}\vec{b}$, and hence $\vec{u} = A^{-1}\vec{b} = \vec{x}$. Thus $\vec{x}$ is the only solution.

5. (a) Suppose A is invertible. We check that A is the inverse of A^{-1} ; namely, $AA^{-1} = I_n$ and $A^{-1}A = I_n$. Thus the inverse of A^{-1} is A ; in symbols $(A^{-1})^{-1} = A$.
- (b) Again suppose A and B are both invertible matrices. We check that $B^{-1}A^{-1}$ is the inverse of AB : $AB B^{-1}A^{-1} = I_n$ and $B^{-1}A^{-1}AB = I_n$. Thus by the definition of inverse matrix, $B^{-1}A^{-1}$ is the inverse of AB ; i.e. $(AB)^{-1} = B^{-1}A^{-1}$.
- (c) Suppose A is an invertible matrix. We want to prove that the inverse of A^T is $(A^{-1})^T$. Again we check the conditions in the definition of matrix inverse: $A^T(A^{-1})^T = (A^{-1}A)^T = (I_n)^T = I_n$ and $(A^{-1})^T A^T = (AA^{-1})^T = (I_n)^T = I_n$ where we used $(AB)^T = B^T A^T$. Thus $(A^T)^{-1} = (A^{-1})^T$.